

WEEK 8

Aim:

To develop a SkillNest web application using HTML, CSS, JavaScript, Node.js, and Express.js with Home, Login, and Registration pages, and to manage user information using a JSON file.

Description:

This experiment demonstrates the development of a basic full-stack web application using Node.js and Express.js. The application consists of Home, Login, and Registration pages created using HTML and CSS. Express.js is used to create the backend server and handle HTTP requests. User registration details are received from the frontend and stored in a JSON file. During login, the entered username and password are compared with the user details stored in the JSON file. The application demonstrates the connection between frontend pages, Express.js backend, and JSON-based data storage.

Program:

SkillNest Full-Stack Web Application Using Node.js and Express.js

Program Code:

```
const express = require("express");
```

```
const fs = require("fs");
```

```
const app = express();
```

```
app.use(express.json());
```

```

    app.use(express.static("public"));

    app.post("/register", (req, res) => {

        let users = JSON.parse(fs.readFileSync("users.json"));

        let user = req.body;

        users.push(user);

        fs.writeFileSync("users.json", JSON.stringify(users, null, 2));

        res.json({

            message: "Registration successful"

        });

    });

    app.post("/login", (req, res) => {

        let users = JSON.parse(fs.readFileSync("users.json"));

        let username = req.body.username;

        let password = req.body.password;

        let user = users.find(

            user => user.username === username && user.password ===
password

        );

        if(user){

            res.json({

```

```
        success: true,  
        message: "Login successful"  
    });  
}  
else{  
    res.json({  
        success: false,  
        message: "User not registered"  
    });  
}  
});
```

```
app.listen(3000, () => {  
    console.log("Server running at http://localhost:3000");  
});
```

Frontend Files:

- [index.html](#) – Home page

Code :

```
<html>  
  
  <head>  
  
    <title>Home-Page</title>
```

```
<style>

    body{

        background-color: cyan;

        background-image: url("./bgimage2.jpg");

    }

    .text1 {

        flex:0.9;

        background-color:palevioletred;

        margin-top: 50px;

        align-content: center;

        border-radius: 35px;

        display: flex;

        flex-direction: column;

        gap:25px;

        align-items: center;

    }

    .items{

        display: flex;

        flex-direction: row;

        justify-content: space-around;

    }


```

```
button{  
    height: 50px;  
    width: 100px;  
    background-color: darkblue;  
    -webkit-text-fill-color: aliceblue;  
}  
button a{  
    text-decoration: none;  
    -webkit-text-fill-color: aliceblue;  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h1 style="text-align: center;">Welcome To SkillNest <p  
style="font-size: 20px ;font-style:italic"> ~ Learn,Build and Grow  
</p></h1>
```

```
<div class="items">
```

```

```

```

<div class="text1">

    <p style="font-style: normal; font-size: 18px;line-height: 2;">

        <b>Welcome to SkillNest, a simple and user-friendly learning
platform designed for students who want to explore new technologies and
improve their skills. Discover useful learning resources, explore different
areas of technology, practice what you learn through hands-on activities, and
build your confidence one skill at a time. Whether you are a beginner or
looking to strengthen your existing knowledge, SkillNest provides a
convenient space to learn, practice, and grow at your own pace.

    </b></p> <br><br>

        <button><a href="register.html">Register</a></button>

    <br>

    <p><b>Already a user ? <a href="login.html">Login</a></b></p>

</div>

</div>

</body>

</html>

```

- [login.html](#) – Login page

Code :

```

<html>

<head>

    <title>Login-Page</title>

```

```
<meta name="viewport" content="width=device-width">
```

```
<meta charset="UTF-8">
```

```
<style>
```

```
body{
```

```
    background-color: cyan;
```

```
    background-image: url("./bgimage2.jpg");
```

```
}
```

```
.heading{
```

```
    color: white;
```

```
    text-align: center;
```

```
    padding-top: 20px;
```

```
}
```

```
.container{
```

```
    margin-left: auto;
```

```
    margin-right: auto;
```

```
    height: 300px;
```

```
width: 300px;

background-color: lavender;

box-shadow: 10px 10px 20px grey;

border-radius: 20px;

display: flex;

flex-direction: column;

margin-top: 80px;

}
```

```
.container h3{

    align-self: center;

}
```

```
label{

    margin-top: 20px;

    color: darkblue;

    margin-left: 10%;

    margin-right: 10%;

}
```

```
input{
```

```
display: block;
margin-left: 10%;
margin-right: 10%;
height: 30px;
}
```

```
#btn{
background-color: darkblue;
color: white;
width: 80%;
align-self: center;
margin-top: 30px;
height: 30px;
}
```

```
.register{
text-align: center;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h1 class="heading">SkillNest</h1>
```

```
<div class="container">
```

```
<h3>User Login</h3>
```

```
<label>Username</label>
```

```
<input type="text" id="name1" placeholder="Enter your username">
```

```
<label>Password</label>
```

```
<input type="password" id="password" placeholder="Enter your  
Password">
```

```
<button onclick="loginUser()" id="btn">Login</button>
```

```
<p class="register">
```

```
  New user?
```

```
  <a href="register.html">Register</a>
```

```
</p>
```

```
</div>
```

```
<script>
```

```
function loginUser(){

    let username = document.getElementById("name1").value;
    let password = document.getElementById("password").value;

    fetch("/login", {
method: "POST",
headers: {
        "Content-Type": "application/json"},
body: JSON.stringify({
        username: username,
        password: password
    })
})

.then(response => response.json()

    .then(data => {
```

```
alert(data.message);

if(data.success){

    window.location.href = "success.html";

    } });

}

</script>

</body>

</html>
```

- [register.html](#) – Registration page

```
<html>

<head>

    <title>Registration-Page</title>

    <meta name="viewport" content="width=device-width">

    <meta charset="UTF-8">

<style>

    body{

        background-color: cyan;

        background-image: url("./bgimage2.jpg");

    }
```

```
.heading{  
    color: black;  
    text-align: center;  
    padding-top: 20px;  
}
```

```
.container{  
    margin-left: auto;  
    margin-right: auto;  
    width: 350px;  
    background-color: lavender;  
    box-shadow: 10px 10px 20px grey;  
    border-radius: 20px;  
    display: flex;  
    flex-direction: column;  
    margin-top: 30px;  
    padding: 20px;  
}
```

```
.container h2{
```

```
    text-align: center;

    color: darkblue;
}
```

```
label{

    margin-top: 12px;

    color: darkblue;
}
```

```
input, select{

    height: 30px;

    margin-top: 5px;
}
```

```
.gender{

    display: flex;

    gap: 15px;

    margin-top: 5px;
}
```

```
.gender input{
```

```
    height: auto;
}
```

```
#btn{
    background-color: darkblue;
    color: white;
    height: 35px;
    margin-top: 20px;
    border: none;
    border-radius: 5px;
}
```

```
.login{
    text-align: center;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h1 class="heading">SkillNest Registration</h1>
```

```
<form class="container" id="registerForm">
```

```
<h2>Create Account</h2>
```

```
<label>Full Name</label>
```

```
<input type="text" name="fullname" placeholder="Enter your full  
name">
```

```
<label>Username</label>
```

```
<input type="text" name="username" placeholder="Enter your  
username">
```

```
<label>Email</label>
```

```
<input type="email" name="email" placeholder="Enter your email">
```

```
<label>Password</label>
```

```
<input type="password" name="password" placeholder="Enter your  
password">
```

```
<label>Confirm Password</label>
```

```
<input type="password" name="confirmPassword"  
placeholder="Confirm your password">
```

<label>Gender</label>

<div class="gender">

<label>

<input type="radio" name="gender" value="Male"> Male

</label>

<label>

<input type="radio" name="gender" value="Female"> Female

</label>

<label>

<input type="radio" name="gender" value="Other"> Other

</label>

</div>

<label>Date of Birth</label>

<input type="date" name="dob">

<label>Select Skill</label>

```
<select name="skill">
  <option value="">Select a skill</option>
  <option value="Programming">Programming</option>
  <option value="Web Development">Web Development</option>
  <option value="Data Science">Data Science</option>
  <option value="Artificial Intelligence">Artificial
Intelligence</option>
  <option value="Database Management">Database
Management</option>
</select>
```

```
<button type="submit" id="btn">Register</button>
```

```
<p class="login">
```

```
  Already a user?
```

```
  <a href="login.html">Login</a>
```

```
</p>
```

```
</form>
```

```
<script>
```

```
document.getElementById("registerForm").addEventListener("submit",
function(event){
```

```
event.preventDefault();

let formData = new FormData(this);

let user = Object.fromEntries(formData);

fetch("/register", {
  method: "POST",
  headers: {
    "Content-Type": "application/json"
  },
  body: JSON.stringify(user)
})

.then(response => response.json

.then(data => {
  alert(data.message);

  if(data.success){
    window.location.href = "login.html";
  }

});

});

</script>

</body>

</html>
```

- [success.html](#) – Successful login/registration page

Code:

```
<html>

<head>

  <title>Success-Page</title>

  <meta name="viewport" content="width=device-width">

  <meta charset="UTF-8">

  <style>

    body{

      background-color: cyan;

      background-image: url("../bgimage2.jpg");

    }

    .container{

      width: 400px;

      height: 250px;

      background-color: lavender;

      margin: 150px auto;

      border-radius: 20px;

      box-shadow: 10px 10px 20px grey;
```

```
display: flex;

flex-direction: column;

align-items: center;

justify-content: center;

}
```

```
h1{

    color: darkblue;

}
```

```
p{

    font-size: 18px;

}
```

```
a{

    text-decoration: none;

    background-color: darkblue;

    color: white;

    padding: 10px 25px;

    border-radius: 5px;

}
```

```
</style>

</head>

<body>

  <div class="container">

    <h1>Login Successful!</h1>

    <p>Welcome to SkillNest.</p>

    <a href="index.html">Go to Home</a>

  </div>

</body>

</html>
```

Data: users.json

Initially, the JSON file contains an empty array:

```
[]
```

After successful registration, the submitted user details are stored in the JSON file in the following format:

```
[
  {
    "id": 1,
    "name": "Lasya",
    "username": "lasya123",
```

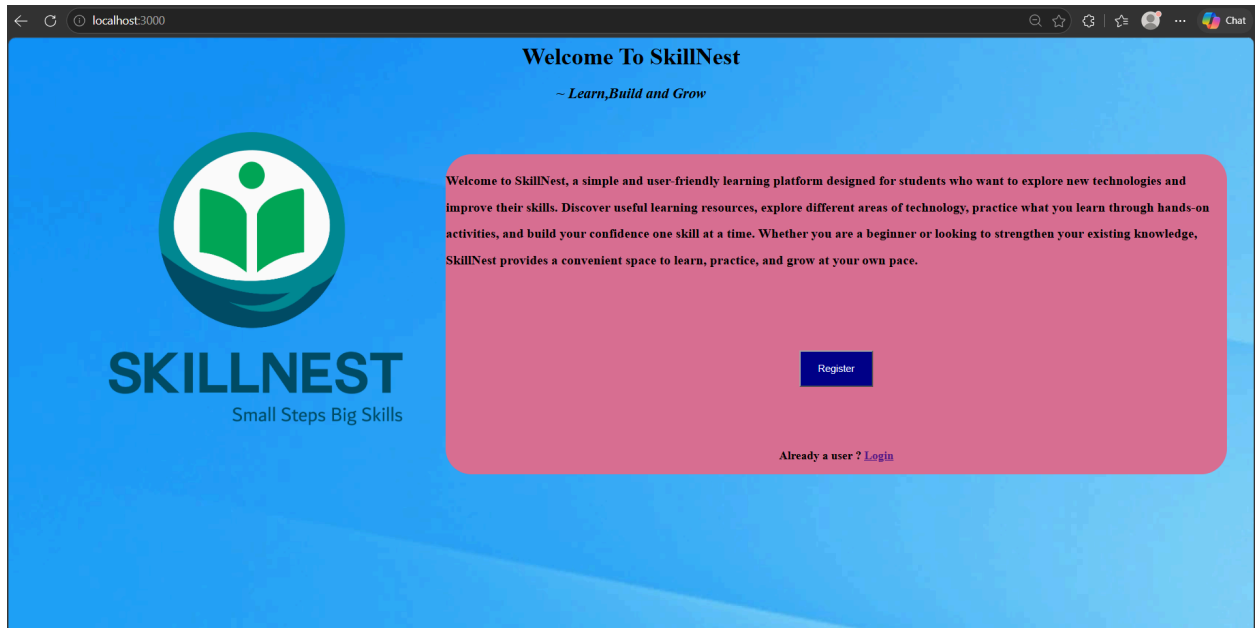
```
    "email": "lasya@gmail.com",  
    "password": "1234"  
  }  
]
```

Execution of the Program

Method	URL/Route	Purpose
GET	http://localhost:3000	Displays the SkillNest Home page
GET	http://localhost:3000/login.html	Displays the Login page
GET	http://localhost:3000/register.html	Displays the Registration page
POST	/register	Receives registration details and stores them in users.json
POST	/login	Checks username and password from users.json

Output:

Home Page:



Registration Page:

The screenshot shows the registration page of the SkillNest application. The browser's address bar indicates the URL is `localhost:3000/register.html`. The page has a blue background with a subtle geometric pattern. In the center, there is a light purple rounded rectangle titled "SkillNest Registration" and "Create Account". The form contains the following fields and options: "Full Name" (text input with "Behara Lasya sree"), "Username" (text input with "lasya"), "Email" (text input with "lasya6086@gmail.com"), "Password" (text input with placeholder "Enter your password"), "Confirm Password" (text input with placeholder "Confirm your password"), "Gender" (radio buttons for "Male", "Female", and "Other"), "Date of Birth" (date input with "22 - 06 - 2006"), and "Select Skill" (dropdown menu with "Select a skill"). At the bottom of the form is a dark blue "Register" button. Below the button, the text "Already a user? [Login](#)" is displayed.

Registration:

localhost:3000 says
Registration successful

OK

Full Name
Behara Lasya

Username
lasya_sree

Email
lasya6986@gmail.com

Password

Confirm Password

Gender
☐ Male ☒ Female ☐ Other

Date of Birth
22 - 06 - 2006

Select Skill
Artificial Intelligence

Register

Already a user? [Login](#)

Login Page:

SkillNest

User Login

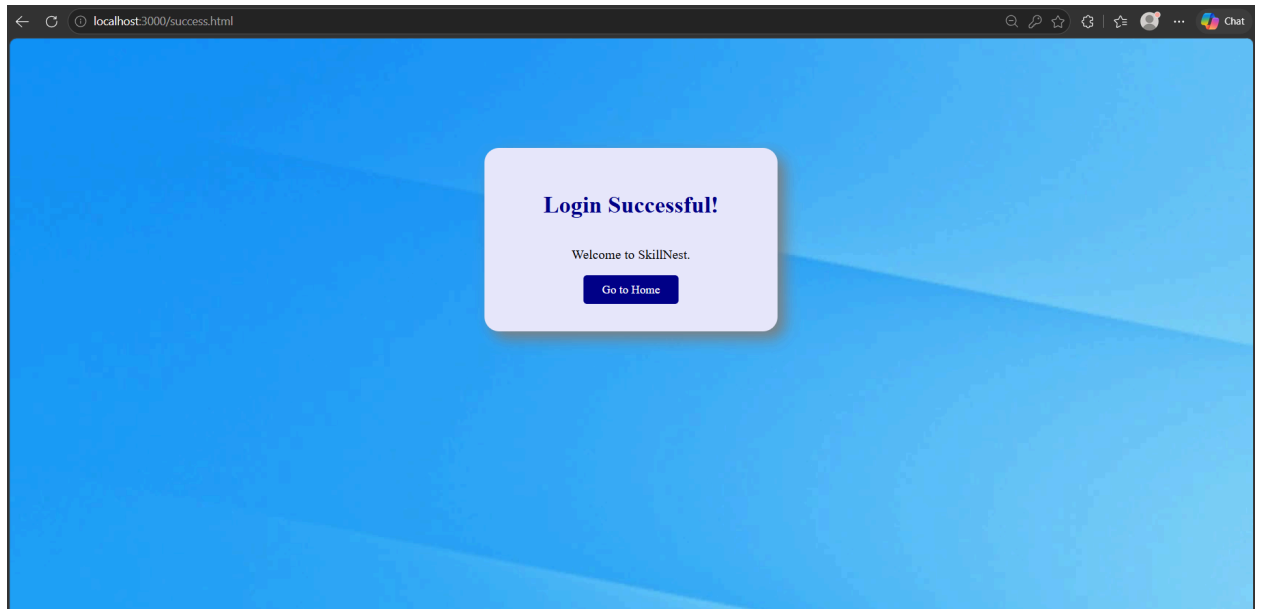
Username
lasya

Password

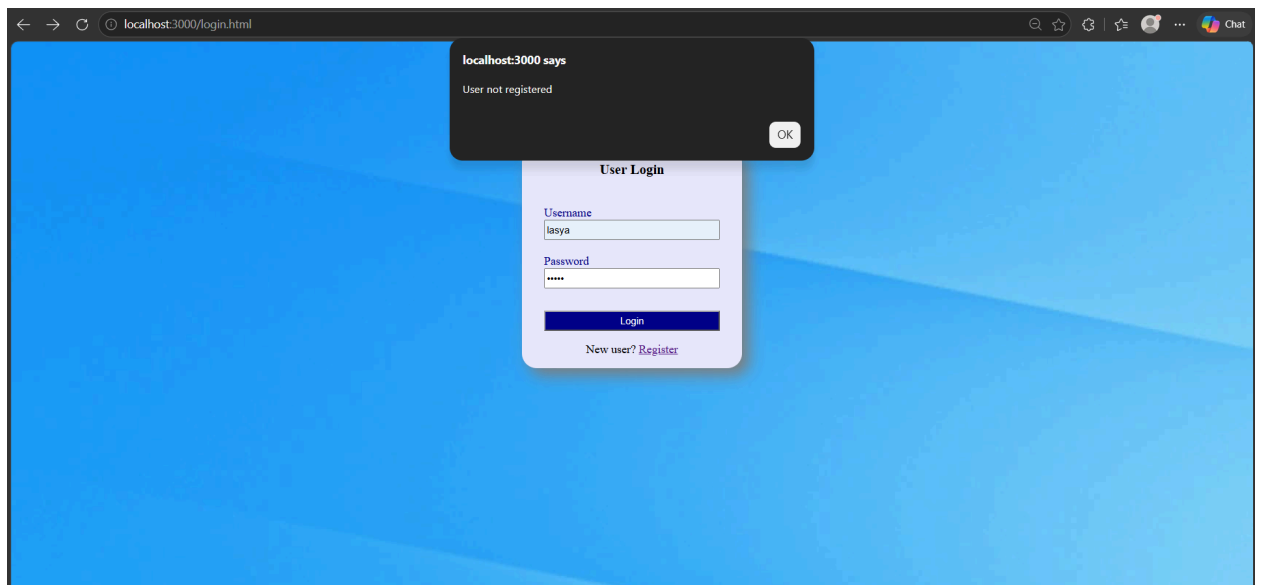
Login

New user? [Register](#)

Successful Login:



Invalid Login:



Result:

The SkillNest full-stack web application was successfully developed using HTML, CSS, JavaScript, Node.js, and Express.js. The application demonstrates frontend page creation, Express.js routing, HTTP GET and POST methods, form handling, JSON file operations, user registration, and login authentication using JSON-based user data.